

MURPHY DIGGINS

Biomedical Engineering Student

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EDUCATION

University of Wisconsin–Madison | Madison, WI

B.S. Biomedical Engineering, Biomaterials Track | GPA: 3.50/4.00 | Dec. 2027

Coursework: Biomaterials, Biomechanics, Tissue Engineering, Microscopy, Design & Leadership, Organic Chemistry

Involvement: Phi Delta Theta - Member/Judicial Board Member, Biomedical Engineering Society, IFC Softball, Party of Poets, Humorology

EXPERIENCE

Franck Laboratory, University of Wisconsin–Madison | Madison, WI

Undergraduate Research Assistant | May 2026 – Present

- Investigate mechanical strain and effects of substrate stiffness on microglial activation using primary microglial cell culture, immunocytochemistry, fluorescence microscopy, and ImageJ/Fiji.
- Fabricate PDMS substrates and design experimental fixtures using CAD and 3D printing to improve experimental reproducibility and workflow.

Pfleger Laboratory, University of Wisconsin–Madison | Madison, WI

Undergraduate Research Assistant | Mar. 2026 – Aug. 2026

- Developed polymer-based biomimetic root models and conducted bacterial adhesion and biofilm assays to study plant-microbe interactions.
- Designed and 3D-printed custom molds and experimental hardware while optimizing material formulations and fabrication processes.

Additional Experience | Various Employers | 2021-2025

Park Ranger • Paraprofessional • Pool Technician • Golf Course Guest Services & Grounds Crew

PROJECTS

Wearable Thermal Management Device | Independent Project

- Designed a sensor-controlled cooling device and validated performance at 77°C ambient temperature while maintaining <15°C surface temperatures.
- Developed a Bluetooth mobile app for live graphing, three-sensor temperature monitoring, data logging, and historical analysis.

Bone Graft Biomaterials & Bioreactor Design

- Led biomaterial selection for a bone graft system and contributed to bioreactor design and electronics integration.
- Designed a custom sample holder in SolidWorks and manufactured it using manual lathe and mill machining.

TECHNICAL SKILLS

Lab: Mammalian & Bacterial Cell Culture, Immunocytochemistry, Sterile Technique, Biological Safety Cabinet (BSC), Fluorescence Microscopy, ImageJ/Fiji, PDMS, Biofilm Assays

Engineering: SolidWorks, Fusion 360, CAD, 3D Printing, Rapid Prototyping, Lathe & Mill, Experimental Design

Software/Electronics: MATLAB, Arduino IDE, Bluetooth/BLE, Sensor Integration, PreForm